

ECOTRACER

A user's guide to tracking contaminants using the Ecopath with Ecosim (EwE) approach



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Date : 1 August 2020
Document version 1.0
EwE software version 6.6.3



**Ecopath
International
Initiative**

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DE RADIOPROTECTION
ET DE SÛRETÉ NUCLÉAIRE

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ABSTRACT

Ecotracer is a useful tool within the Ecopath and Ecosim (EwE) modelling approach to track radioisotopes, contaminants, persistent pollutants, or stable isotopes through a food web model. After achieving a mass-balanced Ecopath model, Ecotracer can be used with the Ecosim (time dynamic) or Ecospace (spatial-temporal dynamic) to track the flow of the pollutant through the modelled ecosystem.

EwE uses a mass-balance approach, and the flows of a contaminant due to predator/prey interactions are tracked within the underlying Ecopath model. However, Ecotracer also needs parameters for groups based on a kinetic toxicology approach to estimate initial conditions. However, similar to Ecopath, Ecotracer can become dynamic through either the use of Ecosim or Ecospace to follow the changes of a contaminant that has different temporal inputs or to variations in temporal spatial concentrations in the water column.

The purpose of this manual is to give a full description of the Ecotracer approach including the dynamic equations that describe the basis for the input parameters, and to familiarize users with the various interfaces for the inputs and outputs. Simulation scenarios are also given to allow users to become more familiar with Ecotracer, and instructions are given on how to navigate through the different interfaces used.

INTRODUCTION

Ecotracer is a sub-routine in the Ecopath with Ecosim (EwE) modeling framework (Polovina 1984; Christensen & Walters 2004; Walters and Christensen 2018) that allows the user to follow a contaminant or stable isotope in modelled functional groups and the environment in a balanced Ecopath model. Many EwE models have been made that focus on fishery-related questions, but here the focus is on how the Ecotracer routine is used within EwE to trace contaminants such as radionuclides through an aquatic ecosystem. EwE consists of three routines: Ecopath which is a mass balance interpretation of an ecosystem where, in essence, the production of a group in the model is equal to its consumption; Ecosim allows the user to build in time dynamics to the Ecopath model for events such as changes in contaminant loading to an ecosystem; and Ecospace which allows for the spatial-temporal resolution for such events as the effects of change in loading on marine organisms that result from organisms inhabiting different spatial areas or habitat types that have different environmental concentrations through time.

Typical applications of Ecotracer have been for contaminants such as mercury (Booth and Zeller 2005), ^{14}C (Sandberg 2007; Tierney et al. 2017), and PCBs (Booth et al. 2016) that can have detrimental impacts on human and environmental health. The use of Ecotracer can help to estimate the amount of contaminant or concentration in a group/species of interest, spatial differences in concentration within the same functional groups if Ecospace is used, the fluxes between groups due to trophic interactions, and the importance of diet versus direct environmental uptake. It can also help to estimate functional groups' concentrations when such data are lacking (i.e., have a starting value of zero) and make forward projections based on changing environmental concentrations. Concentration levels are an important aspect for environmental and human health as, in conjunction with consumption rates, they determine exposure levels that may have detrimental effects. Regulatory limits on the concentration in aquatic products destined for human consumption may also affect trade and fisheries opportunities.

Ecotracer requires a balanced Ecopath model to follow the contaminant, radioisotopes or stable isotopes in the model groups, and environment (e.g., water concentration). Ecotracer when used with Ecosim can provide estimates to important ecotoxicological questions such as: 1) what could be the expected group concentrations if the environmental concentration did not change?; 2) what could be the expected results in group concentrations if the environmental concentration changed through time?; and 3) is there an effect on concentration levels as a result of changing underlying Ecopath parameters such as fishing mortality? The first question is useful if many functional groups in the model lack concentration data. The second question can be important to estimate resulting concentrations in biota if the input into the environment changes. The third can help to understand contaminant flows as a result of changes in the dynamics of the underlying structure of an ecosystem.

Ecotracer when used with Ecospace can help to answer whether there are differences in the same species that occur over a large geographic area, and whether different environmental concentrations in different areas have an affect on the resulting concentrations in organisms. In this case, a two-dimensional representation of the model area is made which has a user

defined spatial resolution (i.e., grid cells). Spatial environmental concentrations can be driven by effluents being released as a point source, or from atmospheric deposits that change over space and time. Effluents released from a point source would be affected by currents resulting in different spatial and temporal distributions, and atmospheric releases could be affected by different levels of releases due to industrial activity through time or accident scenarios as well as currents. In the case of large-scale accidents, such as the Dai-ichi nuclear accident at Fukushima, spatial differences can result from both point sources and differing atmospheric deposits both of which occurred through time.

ECOTRACER DYNAMICS

Ecotracer simulates the contaminant fluxes and resulting amounts and concentrations using a modified transfer contaminant model (e.g., Landrum et al. 1992; Thomann 1981), and applies it to both the environment and biota. Resulting changes at any time step are dependent upon the gains and losses in functional groups and are described in Walters and Christensen (2018) as,

$$\frac{dA_i(t)}{dt} = \alpha_i - \beta_i A_i(t) \quad \text{Eq. 1}$$

where α_i represents the gains ($\text{Bq}\cdot\text{year}^{-1}$) in each functional group i , β_i represents the rate losses (year^{-1}) to each functional group, and A_{io} represents the amount (e.g., Bq) in each functional group i . This general formulation allows different measurement units of substances (e.g., Bq or μg) to be tracked in the modelled environment, and the resulting concentrations (e.g., $\text{Bq}\cdot\text{t}^{-1}$) are computed separately using the biomass output in Ecopath and Ecosim.

The environmental compartment concentrations are also calculated by tracking the gains and losses in the cells representing the environment,

$$\frac{dC_o(t)}{dt} = \alpha_o - \beta_o C_o(t) \quad \text{Eq. 2}$$

where C_o represents the environmental concentration (e.g., $\text{Bq}\cdot\text{km}^{-2}$), α_o represents the gains ($\text{Bq}\cdot\text{year}^{-1}$) in each environmental cell o , and β_o represents the rate losses (year^{-1}) in each cell.

Environment

Gains in the environment originate from the release of contaminants into the environment as a base inflow rate, and from the excretion from organisms. Losses originate from the direct uptake from the environment by organisms, physical decay rates, and base volume exchange. In Ecospace, the environment can be represented by multiple grid cells and thus the gains and losses can be considered to be for each environmental compartment o , such that,

$$\alpha_o = BI_o + \sum_i^n m_i A_i \quad \text{Eq. 3}$$

where Bl_o is the base inflow rate ($Bq \cdot km^{-2} \cdot year^{-1}$) to a grid cell, and $m_i A_i$ are the excretory products for each functional group within each grid cell.

Losses from the environment are due to biological, physical decay processes, environmental volume changes, and direct uptake by organisms, such that,

$$\beta_o C_o = (d_i + V_i) C_o + \sum_i^n u_i B_i C_o \quad Eq. 4$$

where d_i represents the physical decay rate ($year^{-1}$), V_i represents the base volume exchange loss ($year^{-1}$), and the second term ($u_i B_i C_o$) represents the total uptake rate by all functional groups (see below). Temporal changes to the environmental concentration (C_o) can be made by applying a forcing function to the base inflow rate, through a contaminant concentration driver file (Table 1), or by current/advection fields.

Biota

In biota, intake amounts (e.g., $Bq \cdot year^{-1}$) result from direct uptake rates (i.e., respiration) the fraction retained from trophic interactions (i.e., diet), and immigration. i.e.,

$$\alpha_i = u_i B_i C_o + AE_i \sum_{j=prey} Q_{ji} * \frac{A_j}{B_j} + c_i I_i \quad Eq. 5$$

where, C_o represents the environmental concentration ($Bq \cdot km^{-2}$), B_i is the biomass (t) of group i , u_i represent the intake/biomass/environmental concentration/year ($km^2 \cdot t^{-1} \cdot year^{-1}$); AE_i is the assimilation efficiency for each group, Q_{ji} is the consumption rate ($t \cdot year^{-1}$) of group j by group i , A_j is the amount of substance in a group (e.g., Bq), B_j is the prey biomass of each prey item j ($Bq \cdot t^{-1}$); c_i is the group biomass concentration ($Bq \cdot t^{-1}$) and I is the immigrating biomass ($t \cdot year^{-1}$).

The losses from a group ($\beta_i C_i$) are attributed to predation, fisheries, other mortality, excretion and decay, i.e.,

$$\beta_i C_i = \left(\sum_{j=pred} \left(\frac{Q_{ij}}{B_i} \right) + F_i + MO_i + E_i + m_i + d_i \right) C_i \quad Eq. 6$$

where Q_{ij} is the rate of consumption ($t \cdot year^{-1}$) of group i due to predation by j , F_i is the fishing mortality rate ($year^{-1}$), MO_i ($year^{-1}$) is other mortality rate (i.e., non-predation mortality), E_i is the emigrating biomass rate ($year^{-1}$), m_i ($year^{-1}$) is the excretion and/or metabolic rate, and d_i ($year^{-1}$) is the physical decay rate. These rates are multiplied by C_i the amount of contaminant (Bq) in each group i . Excretory products that are released from tissues to the environment are added to the environmental concentration.

The solution for finding the equilibrium amount of contaminant in a primary producer with the resulting concentration only being due to direct uptake, losses to due predation, other mortality, metabolism, and decay is given as,

$$C_{i,eq} = \frac{\mu_i B_i C_o}{\sum_{j=pred} \frac{Q_{ij}}{B_i} + MO_i + m_i + d_i} \quad Eq. 7$$

whereas for other groups an additional term must be accounted for due to the group's prey items; in these cases the equilibrium solution can be defined as,

$$C_{i,eq} = \frac{\mu_i B_i C_o + AE_i \frac{Q_{ji}}{B_i}}{\sum_{j=pred} \frac{Q_{ij}}{B_i} + MO_i + m_i + d_i} \quad Eq. 8$$

The Ecotracer approach is dynamic and extends the basic concentration ratio (CR) approach, but the CR approach is contained within it as,

$$CR_i = \frac{\left(\frac{A_i}{B_i}\right)}{C_o} = \frac{u_i + AE_i \frac{Q_{ji}}{B_i} CR_j}{Z_i + m_i} \quad Eq. 9$$

The amount of contaminant in the detritus compartment originates from the unassimilated consumption resulting from predation, as well as non-predation mortality. Thus, groups feeding on detritus will have exposure levels associated with the contributions from the fraction of unassimilated consumption from all groups. Initial concentrations in the biota and environment are also input parameters that can be used if data is available. For groups lacking contaminant data from field studies or literature data, the model is able to estimate concentration or burdens in the groups, leading to the ability to estimate risk through time and make comparisons to regulatory limits.

ECOTRACER INPUT INTERFACE AND REQUIRED DATA

The Ecotracer interface involves a number of input parameters (e.g., BI) and some that are calculated internally by the program (e.g. gains from diet). Ecotracer's interface can be loaded from the task bar or from the side-menu. The interface is broken into three parts: the side panel (navigator) to choose Ecotracer or one of the other components of EwE; the interface for input for the environment; and input data for the functional groups based on the Ecopath model (Figure 1). The fields for the environment and the functional groups, their description and units are described below.

Navigator

The side panel allows easy access to the different routines within the EwE modelling framework. After creating a balanced Ecopath model, the Ecotracer routine can be selected in this area. Ecotracer parameters must be selected and a name can be given to the scenario. Also, it is necessary to select either 'enable contaminant tracing for Ecosim' if the model is only being run forward in time or 'enable contaminant tracing for Ecospace' if the model is spatialized using the Ecospace. The selection of which is dependent upon the underlying model and whether there are time or spatial-temporal dynamics that are of interest.

Troubleshooting: if you do not get a response in Ecotracer and you have entered a direct absorption rate for primary producers; ensure that Ecosim or Ecospace have the contaminant tracing box checked.

Environment

The simplest case for the environment is if the model area has a near uniform concentration. If this is the case then one concentration can be used for the initial concentration of the environment. If there are differences in time, but the spatial area is treated as being homogenous in concentration, then the base inflow rate can be changed using a csv file that represents either absolute or relative changes to the initial environmental concentration. If there are spatial-temporal variations in environmental contaminants in the model area, then a professional license for EwE is needed. Through the Spatial Temporal Data Framework (STDF; Steenbeek et al., 2013), monthly maps of contaminant concentrations can be inserted into Ecotracer, enabling modellers to incorporate historical and/or projected environmental contaminants into Ecotracer. Brief instructions how to use the STDF with Ecotracer are provided as an annex at the end of this manual.

Initial concentration: concentration in the water. Currently, there is no explicit depth representation in Ecospace and to be consistent with the biomass units (i.e., $\text{t}\cdot\text{km}^{-2}$) in Ecopath the units are expressed per km^2 with the units commonly being mass for most contaminants (e.g., $\text{g}\cdot\text{km}^{-2}$) or becquerels for radioisotopes (i.e., $\text{Bq}\cdot\text{km}^{-2}$). Therefore, units expressed in terms of unit per volume (e.g., $\mu\text{g}\cdot\text{L}^{-1}$) must be converted to the spatial resolution used.

Base inflow rate: the amount entering the environment each year (e.g., $\text{g}\cdot\text{km}^{-2}\cdot\text{year}^{-1}$). Obviously a base inflow rate for a one time release of, for instance, a hazardous chemical would not need a base inflow rate. However, to maintain background levels under equilibrium conditions a base inflow rate is needed to account for uptake and/or physical decay rates of the contaminant.

Base volume exchange loss: the rate at which a contaminant is lost to the environment (year^{-1}).

File for Base Inflow Rate: this is intended to be used when the environment receives a uniform increase. This is a .csv file that is arranged to implement changes to the environment on a relative or absolute scale through time (Table 1).

File for Point source Release: the point source release is another way of implementing changes to the environment on a spatial-temporal scale in conjunction with Ecospace. It requires a series of ASCII files to implement the changes in a spatial cell through time and is a feature of the EwE pro license.

Maximum number of time steps: Ecotracer numerically solves the equations at a higher resolution (i.e., more time steps) than Ecopath, Ecosim and Ecospace, which all have monthly time steps in order to reach a stable solution using an explicit integration method. The number of time steps is important for some contaminants that have relatively high physical decay rates. Therefore, the time steps and the number of calculations are increased to accurately

trace the contaminant within Ecotracer, but Ecotracer integrates itself with the monthly time steps of Ecopath, Ecosim and Ecospace.

Biota

For ease of computation within the Ecotracer program, Ecotracer uses the amount of substance (e.g., Bq or g). However, for output (see below) the user can choose between the amount in a group (e.g., Bq) or biomass concentration (e.g., $\text{g}\cdot\text{t}^{-1}$). When validating model output to field sample data, it is likely that the user would be comparing the biomass concentrations. All functional groups do not need to have starting concentrations, but they should have estimates of direct uptake rates, otherwise the resulting biomass concentrations will only reflect dietary uptakes based on the diet compositions entered for the Ecopath model. While this may not be important for top piscivores due to the fact that their diets mostly determine their contaminant load, it may affect the lower trophic levels of the food web where direct uptake rates may be more important than diet, and are the only entry way for primary producers. Dietary uptake is not entered as part of the Ecotracer routine as it is determined from the diet matrix, the consumption rate, the unassimilated amount, and the proportion of contaminant excreted.

Initial conc: the amount (e.g., Bq) in a group at the start of the model period (i.e., $t = 0$). Groups with no data from sample surveys can have a starting amount of zero.

Concentration in immigrating biomass: the concentration of a contaminant ($\text{Bq}\cdot\text{t}^{-1}$) in a functional group that is entering the model area under consideration.

Direct absorption rate: represents the amount of a contaminant that a functional group takes up from the environment ($\text{km}^2\cdot\text{t}^{-1}\cdot\text{year}^{-1}$). It represents the proportionality coefficient between the direct uptake rates (e.g., $\text{Bq}\cdot\text{year}^{-1}$), and the biomass (t) and environmental concentration (e.g., $\text{Bq}\cdot\text{km}^{-2}$). Generally, the primary producers must have a direct absorption rate, and it is useful to determine the direct absorption rate for other groups otherwise the resulting concentrations will only be due to trophic interactions (i.e., dietary items). Also, direct absorption rates are important for simulating changes in environmental concentrations through time.

Metabolism rate: the rate (year^{-1}) of losses due to the transformation of the contaminant of concern into another form (e.g., methylmercury to ionic mercury). These represent losses to the system.

Decay rate: the instantaneous physical decay rate constant (i.e., λ ; year^{-1}) of radioisotopes and represents losses to the system.

Proportion of contaminant excreted: this is the proportion (0-1) of the contaminant that is excreted back into the environment and stays in the system. It is equivalent to $1 - \text{assimilation efficiency}$.

Excretion rates: excretion rates (year^{-1}) are also described in Walters and Christensen (2018), but at this time are not represented in the user interface. However, if they are small relative

to the surrounding environment they may not be of importance. However, in comparison to the metabolism rate there is an important difference. Excretory products stay within the system (i.e., they are available to be taken up again by biota), whereas the metabolism of contaminants are lost to the system because they are changed into something else.

ECOTRACER OUTPUT INTERFACE AND DATA MANAGEMENT

The interface for the output of the data involves the navigator, a side panel and a graph area (Figure 4). Ecotracer output is selected from the navigator to display the side panel and the graph area. Data on the side panel involves choosing the data that is to be displayed on the graph area, the type of contaminant data displayed, the data displayed on the graphing area, the specific routine run button, and whether or not the data is automatically saved. Once data has been entered into the input interface, the simulation is initiated from the output interface either through the 'Run Ecosim' or 'Run Ecospace' buttons on the side panel to the graphing interface.

The graphical interface displays how the contaminant in the environment and in each functional group changes through time. There is a choice between having the functional groups trace the contaminant by the amount (e.g., Bq) in the group or by the biomass concentration (e.g. $\text{Bq}\cdot\text{t}^{-1}$). The environmental concentration is essentially an area density (e.g., $\text{Bq}\cdot\text{km}^{-2}$). Under an assumption of environmental equilibrium, the environmental concentration on the graph should show a flat line. Under no changes to the functional groups underlying parameters (e.g., fishing mortality), the concentration in each group (e.g., $\text{Bq}\cdot\text{t}^{-1}$) should also be represented by a flat line.

For radioisotopes under assumed equilibrium conditions, the amount lost by radioactive decay would have to be added in as a base inflow rate to maintain equilibrium. Although equilibrium conditions may not be realistic, it is definitely a good way to see if the model is behaving properly, and may be a good assumption if the environmental concentrations do not change much due to a radioisotope having a long half-life, or when long-lived animals (e.g., whales) are not represented in the model as they would not be affected by historical environmental concentrations. The inverse of the P/B ratio can be an indication of the longevity of the groups in the model.

Output files in Ecosim can be saved to a directory by selecting auto-save results. In Ecospace, it is necessary to open Ecospace parameters on the navigator panel and then select the files under the auto-save dialogue box. In most cases, the biomass maps and the contaminant concentration map need to be selected. This is because the concentration map is for the amount and therefore it is necessary to compute the concentration (e.g., $\text{Bq}\cdot\text{t}^{-1}$) from the two sets of data. The path directory for output files can be found in the status window. However, usually for the first run of a scenario it is useful to not save the data and examine the output by hovering over the concentration data on the graph; point-values can be shown by right clicking on a line on the graph and selecting 'show point values'. After adjustments have been made to basic input parameters (Ecopath or Ecotracer) if required, and the point values are in the desired range of expected output values, it is then worthwhile to save the output files.

Currently post-processing of the output files must be done in another program such as Microsoft Excel or R. Models that only use Ecosim for time dynamics will have a single output

file containing a single column of output for each functional group and 12 rows of data for each year with the cells containing the amount in the group. Models using Ecospace have csv output files for each species that cover one time step for one species with the columns representing the longitudinal grid of the map and the rows the latitudinal grid working from the top left cell to the bottom right cell.

Changes in concentration for the environment and functional groups other than those resulting from changes in environmental concentrations result from the kinetic interactions as described earlier. Functional groups with no starting concentrations will have increasing concentrations, and groups with starting concentrations may have changes through time as a result of their losses and gains at each time step. However, differences between a functional group's model output and those with a starting amount based on field collected samples can also be affected by biases in sampling (i.e., sampling not being representative of the population concerned) or by incorrect Ecopath or Ecotracer parameters. The biomass of the group may also change through time most commonly due to trophic interactions or changes in fishing mortalities if these are represented in the model. Thus, the resulting biomass concentrations can be a function of changes in the amount of contaminant in the functional groups and changes in the underlying Ecopath parameters through time.

CONCLUSION

Ecotracer is a useful computer simulation tool to gain knowledge on the transfer of pollutants in an ecosystem. We attempted to give a practical guide to people who are interested in the use of the EwE ecosystem modelling framework a better understanding on the technical use of the routine, and the interfaces used. We hope this inspires people from various backgrounds to engage and gain a better understanding of how contaminants behave in an ecosystem.

ACKNOWLEDGEMENTS

This work has been funded by the Institut de Radioprotection et de Sûreté Nucléaire (IRSN) and the French program Investissement d'Avenir run by the National Research Agency (AMORAD project, grant ANR-11-RSNR-0002, 2013-2022).

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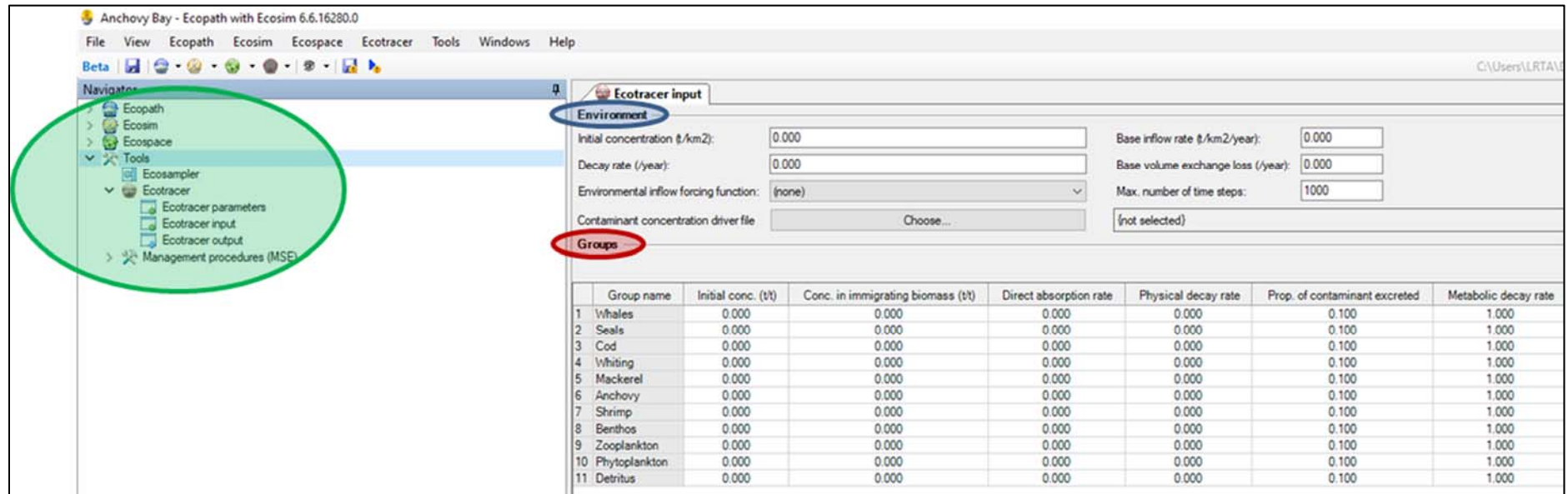


Figure 1. Screen capture of the Ecotracer routine showing the access to the user interface on the side panel (green), the areas for environmental data (blue) and functional group (red) inputs. The input data user interface can also be accessed through the toolbar at the top of the interface.

Start Ecotracer input

Environment

Initial concentration (t/km2): 0.000

Decay rate (/year): 0.000

Environmental inflow forcing function: (none)

Contaminant concentration driver file: Choose...

Base inflow rate (t/km2/year): 0.000

Base volume exchange loss (/year): 0.000

Max. number of time steps: 1000

(not selected)

Clear (X)

Figure 2. Close-up view of the interface for environmental parameters for Ecotracer. Environmental inflow forcing function (highlighted in green) requires a csv file (see Table 1), whereas the contaminant concentration driver file (highlighted in red) uses a series of ASCII files for each grid cell used in Ecospace, and requires the EwE pro license for the spatial-temporal framework.

Groups							
	Group name	Initial conc. (t/t)	Conc. in immigrating biomass (t/t)	Direct absorption rate	Physical decay rate	Prop. of contaminant excreted	Metabolic decay rate
1	Whales	0.000	0.000	0.000	0.000	0.100	1.000
2	Seals	0.000	0.000	0.000	0.000	0.100	1.000
3	Cod	0.000	0.000	0.000	0.000	0.100	1.000
4	Whiting	0.000	0.000	0.000	0.000	0.100	1.000
5	Mackerel	0.000	0.000	0.000	0.000	0.100	1.000
6	Anchovy	0.000	0.000	0.000	0.000	0.100	1.000
7	Shrimp	0.000	0.000	0.000	0.000	0.100	1.000
8	Benthos	0.000	0.000	0.000	0.000	0.100	1.000
9	Zooplankton	0.000	0.000	0.000	0.000	0.100	1.000
10	Phytoplankton	0.000	0.000	0.000	0.000	0.100	1.000
11	Detritus	0.000	0.000	0.000	0.000	0.100	1.000

Figure 3. Close-up view of the interface for functional group parameters for Ecotracer. The direct absorption rate (u_i ; highlighted in red) indicates that primary producers are generally the first compartment that are involved in the transfer of a contaminant through the food web model. The amount taken up by predator/prey interactions are defined by parameters in the underlying Ecopath model and by the diet matrix, and are accounted for internally by EwE. Note that the Initial conc. is the total amount in the functional group (see text for details).

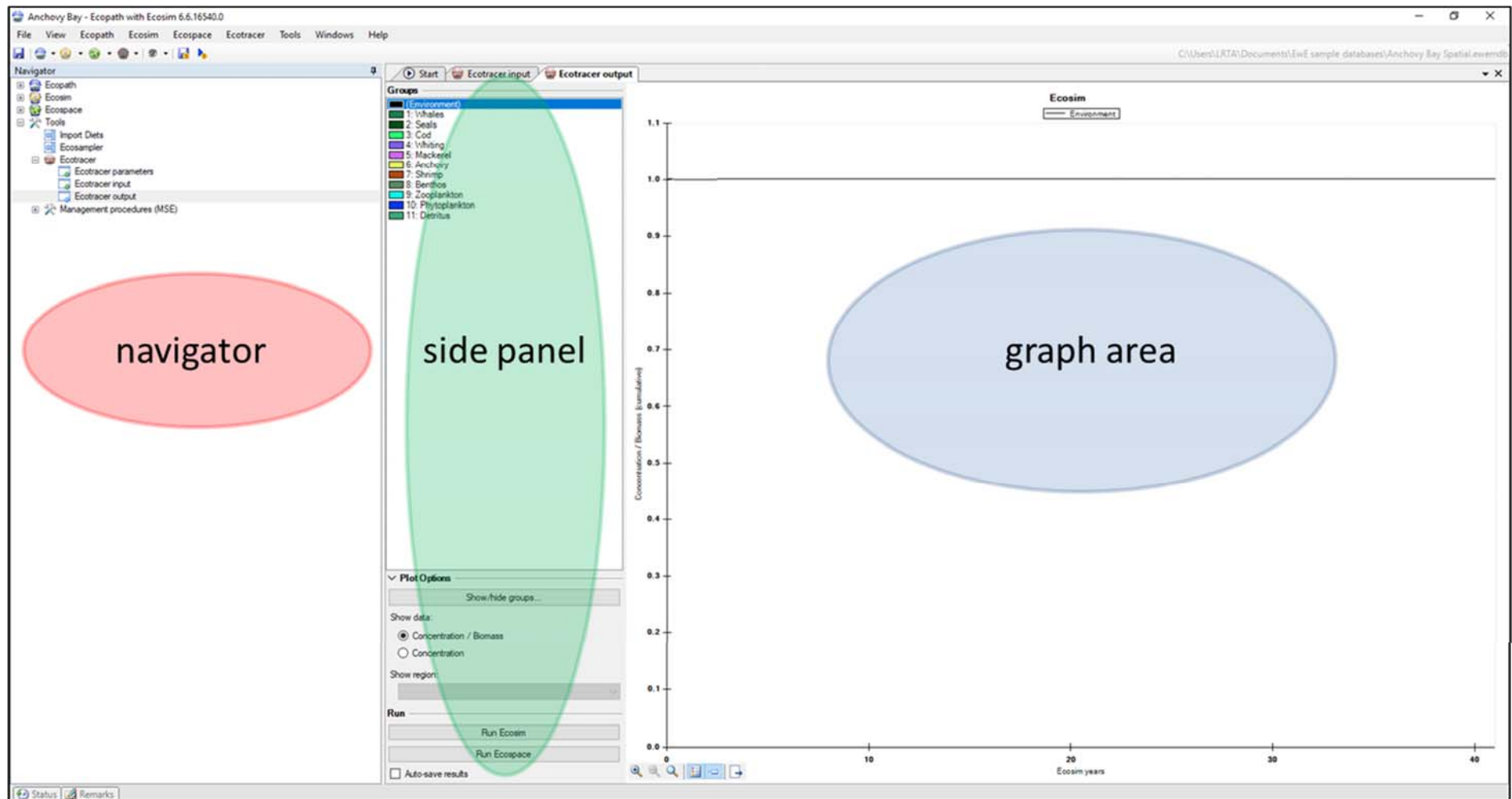


Figure 4. Full view of the Ecotrac output interface composed of the navigator, side panel and graphing area. The graph area, in this example shows that the amount of contaminant in the environment is generally constant through time. It is also possible to select more than one group by using the control or alt key.

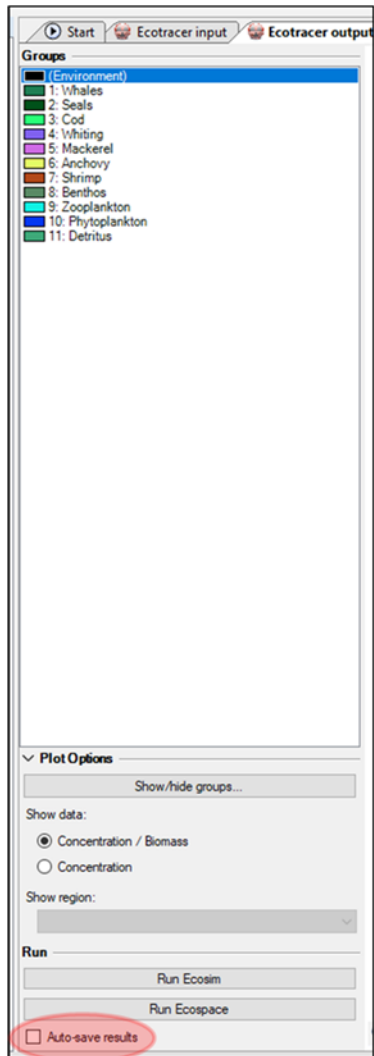


Figure 5. Navigator pane for Ecotracer showing the location of the Status and Remarks area (highlighted in red). These allow the user to determine what changes have been made and to find the directory path for output files generated after a simulation.

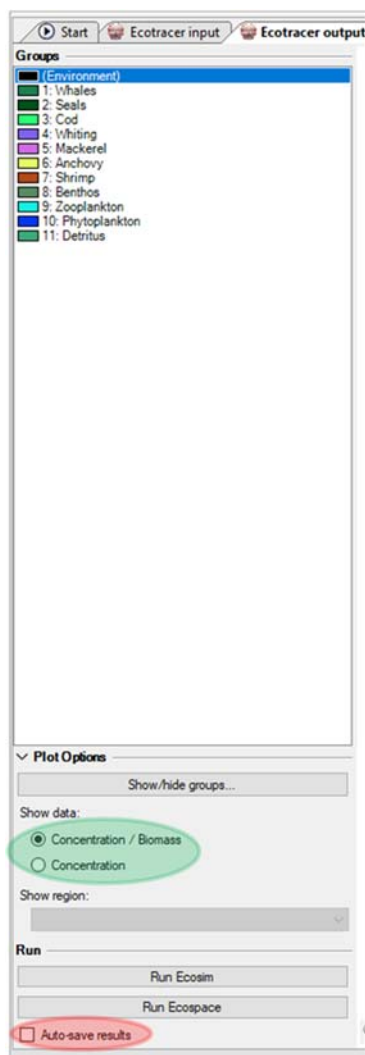


Figure 6. Side panel to the graph area with colour details for the environment and functional groups, plotting data type (highlighted in green) and the box for auto-saving results (highlighted in red) after each simulation run. In the show data section, the label concentration is the amount of contaminant and the concentration/biomass is the amount/biomass (i.e., concentration).

Table 1: Environmental inflow forcing function file that changes the relative base inflow rate from 1 to 3.40 over ten years; pool indicates the environment; and type 2 signifies a relative rate to the original environmental concentrations.

name	Environment
pool	4
type	2
0	1.00
1	1.24
2	1.48
3	1.72
4	1.96
5	2.20
6	2.44
7	2.68
8	2.92
9	3.16
10	3.40

ANNEX I: SIMULATION SCENARIOS (TUTORIALS)

Two simulation scenarios are presented here to give a user a basic introduction to the Ecotracer routine within the EwE modelling framework:

The first represents a simple model with two living groups (phytoplankton and zooplankton) and a detritus group with all necessary data (and could be solved numerically without the use of EwE).

The second is a more complex, and it is necessary to use analysis to fill in the missing data, a situation that is more likely to be come across. In the first example, a tree is given that mimics the navigator panel for ease.

Scenario 1

Scenario 1 consists of three groups with the Ecopath parameter values given (Table 2). As well, zooplankton only eat phytoplankton. For Ecotracer, there is an initial environmental concentration, a direct absorption rate (DAR) for phytoplankton, an assimilation efficiency (AE) for zooplankton, and decay rates are lacking (Table 3). The '>' symbol below describes the direction path and to click, whereas the '=' needs to have data entered including such things as file names that need to be created. In this scenario, it is easy to find the numerical solutions for each group at equilibrium by equations 7 and 8.

Table 2. Ecopath input data for a simple case simulation.

Group	Biomass (t)	P/B (year ⁻¹)	Q/B (year ⁻¹)
Zooplankton	0.0001	1.0	5.0
Phytoplankton	1.0	2.0	-
Detritus	1.0	-	-

Table 3. Ecotracer input data for a simple case simulation.

Group	Initial conc. (Bq·km ⁻²)	DAR (km ⁻² ·t ⁻¹ ·year ⁻¹)	AE (fraction)
Environment	1.0	-	-
Zooplankton	-	-	0.8
Phytoplankton	-	1 x 10 ⁻⁵	-
Detritus	-	-	-

ToolBar

>File

>New model

=choose directory to save the new model and a file name

Navigator panel

>Ecopath

>Input

>Model parameters

=fill in Model year and No. years (5 years)

>Basic input

>Define groups...

=insert two groups and name (phytoplankton and zooplankton); set phytoplankton to producer.00

=ok and save changes

=add data from table 2; save

>Diet composition

=put in diet fraction (zooplankton only eat phytoplankton so 1 for diet fraction)

(Message regarding landings and discards, okay to ignore, there are no fisheries)

> Output

>Basic estimates

>Tools

>Ecotracer

=Name new Ecosim scenario > create

=Name new Ecotracer scenario > create

>Ecotracer parameters

=enable contaminant tracing for Ecosim

>Ecosim

>Input

>Ecosim parameters

= Duration of simulation (years): 5

(notice that the contaminant tracing box is checked)

>Tools

>Ecotracer

>Ecotracer input

=input data from table 3

>Ecotracer output

=choose: show data (concentration/biomass or concentration)

=select all groups to display on graph

> Run Ecosim (results show automatically on graph display)

(note that the environment stays at 1.0, but all the other groups are near zero—there is a colored line along the x-axis; to check the functional group values, select them and they will appear on the graph without the environment concentration)

Scenario 2

This scenario can be carried out in an Excel spreadsheet to solve for the parameters needed for input into Ecotracer. This scenario is based on a more typical situation where there are no reported values for all parameters and it is necessary to make an estimate for the missing parameters. The base Ecopath model is Anchovy Bay, a model that is part of the EwE download and is automatically saved on your computer in the 'EwE sample databases' folder.

Concentration ratios are usually reported in the literature, or have to be derived from separately reported studies for water concentrations and for concentrations in biota. Typically, assuming no temperature, particles (salts), or pressure effects, 1 m³ of water is equal to 1 tonne. Measurements of the contaminant in biota need to be scaled to the biomass unit in Ecopath (i.e., tonnes) as they are usually reported per gram of tissue. Measurements of a contaminant in dry weight should be changed to wet weights using a conversion factor.

Table 4. List of symbols used.

Symbol	Description	Unit
B_i	Biomass	tonne
P/B_i	Production to biomass ratio	year ⁻¹
TL_i	Trophic level	unitless
CR_i	Concentration ratio	unitless
CR_{EwEi}	EwE concentration ratio	unitless
m_i	Excretion rate	year ⁻¹
d_i	Decay rate	year ⁻¹
AE_i	Assimilation efficiency	0 - 1
u_i	Direct absorption rate	km ² ·t·year ⁻¹
$A_{i,eq}$	Equilibrium amount	g
$C_{i,eq}$	Equilibrium concentration	g·t ⁻¹

Starting Information

Ecotracer and all values can be done in a spreadsheet to find missing values. The intent of this scenario is to build a spreadsheet model for all the parameters to run in Ecotracer. The parameters for the Ecopath model (B, P/B, and TL) are taken from the Anchovy Bay model with additional ecotoxicological data being provided. For our purposes we will consider Anchovy Bay to be 1000 m x 1000 m with an average depth of 250 m. ¹³⁷Cs in Anchovy Bay has found to have an activity of 2 Bq·m⁻³.

The following represents a way to estimate the direct absorption rate for groups. Generally, for substances that bioaccumulate, the amount of a substance such as ¹³⁷Cs in a group or species is more dependent on diet than direct uptake at higher trophic levels. However, the direct absorption rate is an important parameter to estimate throughout the food web. Lack of a direct absorption rate at lower trophic levels can lead to an overestimate on the importance of diet or to an error being amplified through the food web with higher trophic levels not reaching measured or likely concentration levels.

Table 5. Starting data for an Ecotracer simulation of ^{137}Cs with data concerning Ecopath parameters (B, TL, and P/B) from the Anchovy Bay model and data representative of what might be measured in field surveys or reported in the literature. The table is designed to replicate an Excel spreadsheet. Values needed to be transferred into the Ecotracer routine include the excretion rate (m_i), physical decay rate (d_i), the amount not assimilated (1-AE), amount of ^{137}Cs (A_i). Methods need to be used to estimate missing values of m_i , A_i , 1-AE, and transforming the environmental concentration from a volume to spatial basis. U_i is solved by finding total gains (TGains) from consumption (Cons) and direct uptake (DU), and total losses (TLoss) from Losses and A_i .

	A	B	C	D	F	G	H	I	J	K	L	M	O	P	Q	R	S	T	U	V	W
1	Group	B	P/B	TL	$\text{Bq}\cdot\text{m}^{-3}$	$\text{Bq}\cdot\text{km}^{-2}$	CR	CR_{EWE}	$\text{Bq}\cdot\text{t}^{-1}$	m_i (year^{-1})	d_i (year^{-1})	Losses (year^{-1})	AE	1-AE	A_i , eq (Bq)	C_i , eq ($\text{Bq}\cdot\text{t}^{-1}$)	Cons	U_i	DU	TGains	TLoss
2	Env	-	-	-	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3	Whales	0.0800	0.050	4.670	-	-	375	-	750	0.023	-	-	-	-	28.0	350.0	-	-	-	-	-
4	Seal	0.0609	0.154	3.362	-	-	112	-	225	3.65	0.023	-	-	-	15.2	250.0	-	-	-	-	-
5	Cod	3.0000	0.310	3.480	-	-	75	-	150	8.76	0.023	-	0.80	0.20	450.0	150.0	-	-	-	-	-
6	Whiting	1.8000	0.581	3.815	-	-	-	-	-	0.023	-	-	-	-	-	-	-	-	-	-	-
7	Mackerel	1.2000	0.723	3.579	-	-	40	-	80	4.38	0.023	-	0.80	0.20	96.0	80.0	-	-	-	-	-
8	Anchov	7.0000	1.149	3.000	-	-	60	-	120	10.95	0.023	-	0.80	0.20	840.0	120.0	-	-	-	-	-
9	Shrimp	0.8000	3.000	3.222	-	-	-	-	-	0.023	-	-	-	-	-	-	-	-	-	-	-
10	Benthos	30.4200	3.000	2.222	-	-	-	-	-	0.023	-	-	-	-	-	-	-	-	-	-	-
11	Zoops	14.8000	35.000	2.000	-	-	25	-	50	23.73	0.023	-	0.75	0.25	740.0	50.0	-	-	-	-	-
12	Algae	9.0000	240.000	1.000	-	-	8	-	16	182.50	0.023	-	1.00	0.00	144.0	16.0	-	-	-	-	-
13	Detr	10.0000	-	1.000	-	-	-	-	-	0.023	-	-	-	-	-	-	-	-	-	-	-

- 1)** The average of $2 \text{ Bq}\cdot\text{m}^{-3}$ needs to be converted to $\text{Bq}\cdot\text{km}^{-2}$. The volume of the ocean is calculated as $250,000,000 \text{ m}^3$ leading to $500,000,000 \text{ Bq}\cdot\text{km}^{-2}$.
- 2)** Elimination rates (K_e ; day^{-1}) need to be determined for groups that lack data. In this scenario $\log K_e$ is plotted against the TL of groups to estimate the K_e for groups that lack data (Figure 7). The resulting relationship is used to estimate the K_e for the groups that lacked data. The K_e is then used to estimate ($K_e \times 365$) the elimination rate (m_i ; year^{-1}).
- 3)** Loss rates in this example result from total mortality (P/B), physical decay rates (d_i), and elimination rates (m_i) and these can be summed to be applied against the amount (A_i) of ^{137}Cs in the groups.
- 4)** Assimilation efficiencies for groups missing data need to be estimated. In this simulation, since all fish groups had an AE of 0.8, and given AEs range between 0.75 and 0.95 a value of 0.8 was used for fish and invertebrate groups without a reported AE. For marine mammals the AE is calculated from the total gains and total losses (i.e., $\text{AE} = \text{Total losses}/\text{Total gains}$).
- 5)** Concentrations ($\text{Bq}\cdot\text{t}^{-1}$) are needed to estimate the direct absorption rate for all groups excluding marine mammals. In this scenario there are 3 groups without concentration data. Therefore, in this simulation a regression analysis is done plotting concentration against trophic level (Figure 8). The importance of estimating a direct absorption rate is for simulations done when there are changing environmental concentrations or changes to the underlying Ecopath input parameters.
- 6)** Calculating the gains from consumption in order to solve for u_i . Recall that at equilibrium when $dC_i(t)/dt = 0$, that the gains are equal to the losses. We have set up the loss rate (step 3), and the gains, excluding marine mammals, are both due to the direct absorption rate (from step 5) and consumption gains. Consumption gains are estimated from AE_i , Q_i (consumption), and A_j/B_j (concentration in diet items of predator i). Unfortunately, the concentration of detritus is not known, but it is derived from all the unassimilated consumption from the Ecopath model. As a result, it is an iterative process that involves an estimate of what the concentration in detritus would be. Generally, a good guess is to start with phytoplankton as it is generally the largest contributor to detritus; in Anchovy Bay, phytoplankton contribute approximately 70 % of the concentration to detritus. Thus, starting detritus with the same concentration as phytoplankton to determine the u_i for all groups, and then run Ecotracer to get the estimate of the concentration of ^{137}Cs for detritus. Then re-run with the new concentration for detritus. Before the second iteration it is necessary to re-calculate the gains from consumption; if the Excel spreadsheet is set up with formulas this will change the values of u_i as well. This example started with a value of $16 \text{ Bq}\cdot\text{t}^{-1}$ for detritus, and after the first run Ecotracer estimated $22.4 \text{ Bq}\cdot\text{t}^{-1}$. Using this value for the second iteration and after changing the consumption values and u_i for the groups the second iteration estimated a value of $21.92 \text{ Bq}\cdot\text{t}^{-1}$ (close enough!).

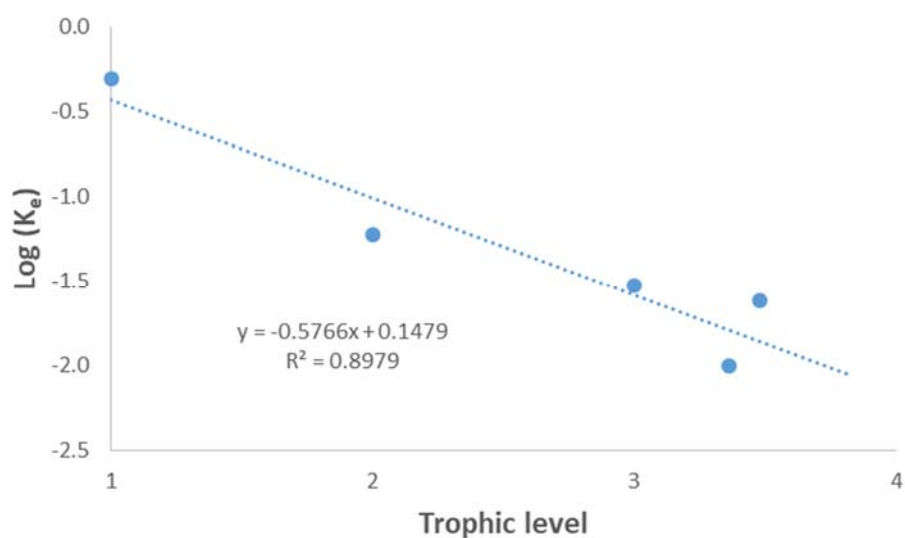


Figure 7. Relation between the elimination rate constant (K_e) and trophic level used to estimate the elimination rate (m_i).

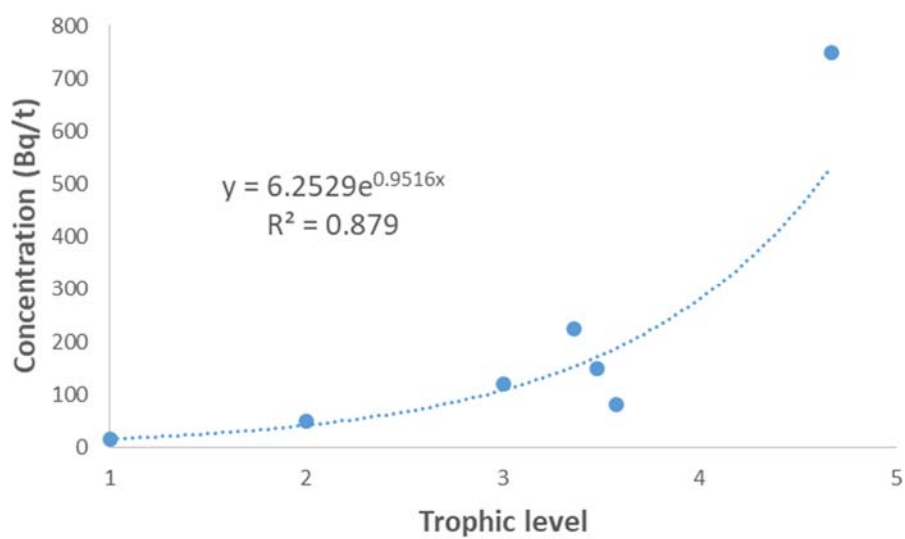


Figure 8. Relation between concentration in functional groups and trophic level used to estimate the direct adsorption rate.

Table 6. Values used for the simulation of ^{137}Cs in Anchovy Bay. Values in **bold** are data taken from the Ecopath model with B = biomass (t), TL (trophic level), P/B (production:biomass ratio; year⁻¹) or that would be representative of field sample measurements or reported literature values (CR = concentration ratio), CR_{EWE} (Ecopath concentration ratio), AE (assimilation efficiency), and Ci,eq (concentration; Bq·t⁻¹). *Italicized* and bold values for the environmental concentration (Bq·km⁻²), m_i (year⁻¹), d_i (physical decay rate; year⁻¹), 1-AE, A_i (Bq), and u_i (km²·t·year⁻¹) need to be entered into the Ecotracer routine. Losses (year⁻¹) are the sum of P/B, m_i and d_i and are applied to A_i to determine total losses (TLoss; Bq·year⁻¹). Cells with no data are estimated by Ecotracer; underlined values are also estimated by Ecotracer but are entered here to show how the estimate of the direct absorption rate u_i is calculated. 1-AE for marine mammals are determined as TLosses/TGains. Total gains result from consumption (Cons; Bq·year⁻¹) and direct uptake (u_i*B_i*C_o; Bq·year⁻¹).

	A	B	C	D	E	F	G	H	I	J	K	L	M	O	P	Q	R	S	T	U
1	Group	B	P/B	TL	Bq·m ⁻³	Bq·km ⁻²	CR	CR _{EWE}	m _i	d _i	Losses	AE	1-AE	A _i , eq	C _i , eq	Cons	u _i	DU	TGains	TLoss
2	Env	-	-	-	2	<i>5x10⁸</i>	-	-	-	0.023	-	-	-	-	-	-	-	-	-	-
3	Whales	0.0800	0.050	4.670	-	-	375	1.5x10 ⁻⁶	<i>1.10</i>	0.023	1.168	-	0.26	60.0	750.0	94.2	-	-	94.2	70.1
4	Seal	0.0609	0.154	3.362	-	-	112	4.5x10 ⁻⁷	3.65	0.023	3.827	-	0.06	13.7	225.0	55.8	-	-	55.8	52.4
5	Cod	3.0000	0.310	3.480	-	-	75	3.0x10 ⁻⁷	8.76	0.023	9.093	0.80	0.20	450.0	150.0	452.5	<i>2.43x10⁻⁶</i>	3639.4	4091.9	4091.9
6	Whiting	1.8000	0.581	3.815	-	-	-	3.8x10 ⁻⁷	<i>3.29</i>	0.023	3.889	0.80	<i>0.20</i>	<u>344.3</u>	<u>191.3</u>	453.2	<i>9.84x10⁻⁷</i>	885.8	1339.0	1339.0
7	Mackerel	1.2000	0.723	3.579	-	-	40	1.6x10 ⁻⁷	<i>4.38</i>	0.023	5.126	0.80	0.20	96.0	80.0	388.9	<i>1.72x10⁻⁷</i>	103.2	492.1	492.1
8	Anchovy	7.0000	1.149	3.000	-	-	60	2.4x10 ⁻⁷	10.95	0.023	12.122	0.80	0.20	840.0	120.0	2556.4	<i>2.18x10⁻⁶</i>	7626.1	10182.5	10182.5
9	Shrimp	0.8000	3.000	3.222	-	-	-	2.4x10 ⁻⁷	<i>7.30</i>	0.023	10.323	0.80	<i>0.20</i>	<u>94.8</u>	<u>118.5</u>	406.1	<i>1.43x10⁻⁶</i>	572.7	978.8	978.8
10	Benthos	30.4200	3.000	2.222	-	-	-	1.1x10 ⁻⁷	<i>27.01</i>	0.023	30.033	0.80	<i>0.20</i>	<u>1608.6</u>	<u>52.9</u>	8049.9	<i>2.65x10⁻⁶</i>	40261.5	48311.4	48311.4
11	Zoops	14.8000	35.000	2.000	-	-	25	1.0x10 ⁻⁷	23.73	0.023	58.748	0.75	0.25	740.0	50.0	25861.0	<i>2.38x10⁻⁶</i>	17612.6	43473.5	43473.5
12	Algae	9.0000	240.000	1.000	-	-	8	3.2x10 ⁻⁸	182.50	0.023	422.523	1.00	0.00	144.0	16.0	-	<i>1.35x10⁻⁵</i>	60843.3	60843.3	60843.3
13	Detr	10.0000	-	1.000	-	-	-	-	-	0.023	-	-	-	-	-	-	-	-	-	-

ANNEX II: DRIVING ENVIRONMENTAL CONTAMINANTS THROUGH THE SPATIAL TEMPORAL DATA FRAMEWORK

Environmental distributions of contaminants can be made time-varying through the spatial temporal framework (STDF). Incorporating monthly changing patterns of contaminant concentrations into Ecotracer is a three-step process: 1) Prepare maps in a GIS; 2) define external data connections in EwE; and 3) connect an external data connection to drive the contents of the Ecospace contaminants layer.

1. Prepare maps in a GIS

Although the STDF contains a GIS engine that converts GIS data to the Ecospace grid (Steenbeek et al. 2013), it is highly recommended that ecosystem modellers perform this step in a dedicated GIS software to minimize GIS processing artifacts.

The step to prepare maps in a GIS entails the necessary steps to convert contaminant distributions over time into discrete, monthly ESRI ASCII maps that match the exact dimensions, cell size, spatial extent, and geospatial projection of the base map in an Ecospace scenario.

The Ecospace base map can be extracted from a loaded Ecospace scenario as follows:

- Click Navigator > Ecospace > Input > Maps
- Double-click the Depth layer at the right-most panel
- In the 'Edit layer' interface that appears, select the Data tab
- In the Data tab, select Export > To ASCII...

The actual steps to convert contaminant distributions to monthly Ecospace maps can require a number of GIS operations, depending on the format and content of the original contaminant data. Operations may require:

- Reprojecting to the Ecospace base map (most likely WGS84 / epsg:4326)
- Temporal clustering
- Clipping
- Kriging (point data) and/or resampling (vector data)
- Exporting to the ESRI ASCII format.

2. Define external data connections in EwE

Once one or more time series of monthly ASCII maps have been prepared, Ecospace needs to know where to find these time series on the local machine. Do the following:

- Click Menu > Ecospace > Define external data connections...
- In the "Define external data connections" interface, select connection type "A series of Ecospace ASCII maps over time" and click Create (Figure 9). This will define the time series, and will bring up the "Configure external data" interface to configure the new time series (Figure 10).

- In the “Configure external data” interface, enter a unique name to identify the time series in the Ecopath with Ecosim user interface.
- It is advisable to use the description to describe where the maps time series came from.
- The variable field serves to limit where the external data connection appears in the Ecopath with Ecosim user interface. For contaminants, leave this field at its default value (No variable).
- Now add the actual maps to the external data connection. Click Browse and navigate the folder that contains the maps. Select all the maps, and click “Open”. The maps will appear in the “Configure external data” interface.
- This interface has limited abilities to parse the date from map file names. Please check the “Time” column and ensure that each map has a correct time tag. One can modify time tags by hand, or with the tools provided in the “Set file times” section.
- Once all maps are correctly time-tagged, close the “Configure external data”. The newly configured time series of external maps will appear in the “Define external data connections” interface.
- Close this interface too.

Ecospace now has the information to locate and integrate time series of external ASCII contaminant maps into the running model.

3. Connect the external data to the Contaminants layer

Now at least one time series of contaminants in seawater has been defined, this time series needs to be connected to drive the contents of the Ecospace contaminants map layer. Do the following:

- In the Navigator, open Ecospace > Input > External data (Figure 11)
- Under input maps, locate the environmental contaminants map. One can also use the on-screen filter options to reduce the number of visible input maps
- In the row for the environmental contaminants map, click the cell “Slot 1” to assign an external connection here. This brings up an interface dedicated to configuring how external data is integrated into Ecospace (Figure 12)
- Here, from the list of available connections, select the connection that you wish to apply, and click the apply (➡) button. The connection now appears under the list of “Connected” external data connections.
- An important aspect is to set a proper scale to ensure that the ¹³⁷Cs concentrations as obtained from external data is properly scaled to the units expected by Ecospace.
- Click Close.
- Run Ecospace.

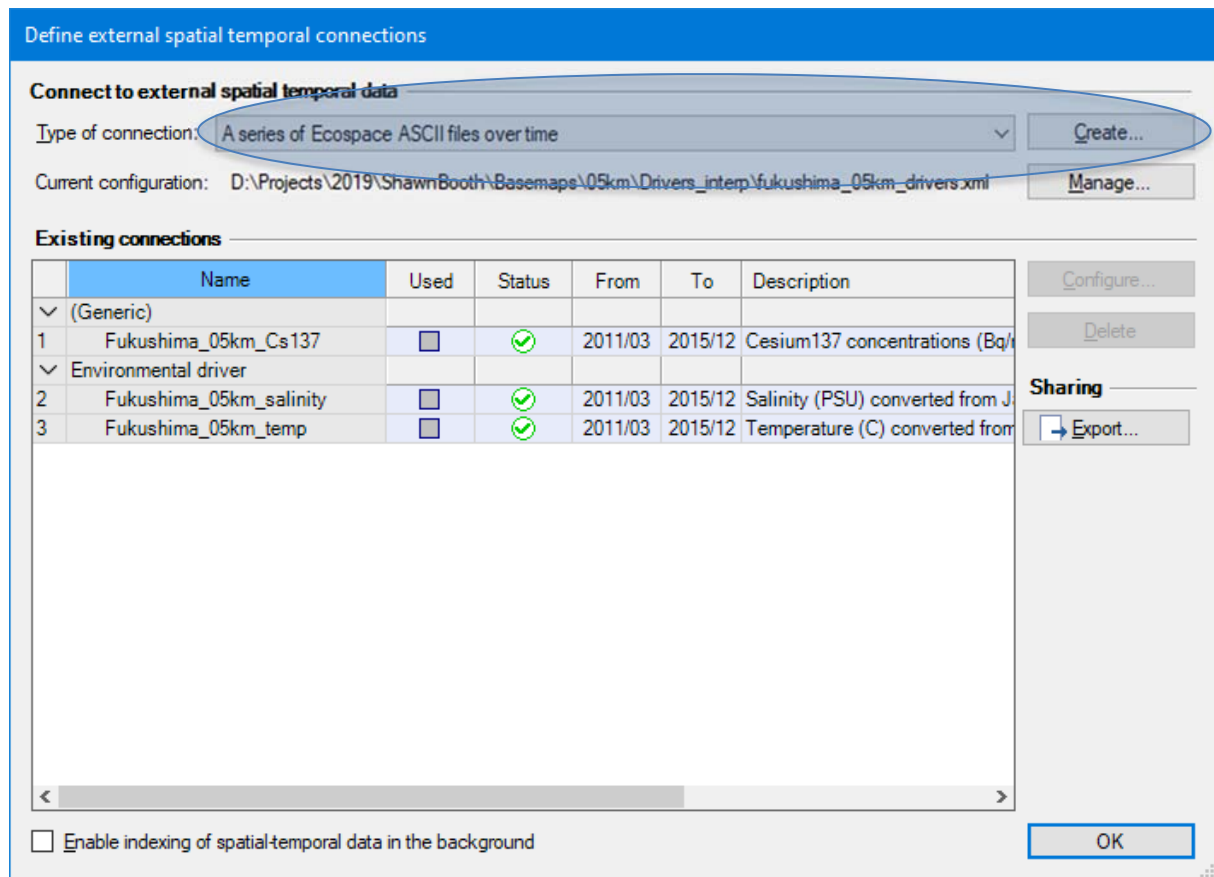


Figure 9: To define a new spatial-temporal data connection, select the type of the connection and click Create. From this interface one can also reconfigure external data connections that were previously created.

Configure external data connection

▼ **Description**

Name: Fukushima_05km_Cs137

Description: Cesium 137 concentrations (Bq/m3) converted from Japanese model at 5km resolution

Variable: (No variable)

Files

Location: D:\Projects\2019\ShawnBooth\Basemaps\...\cs137 Browse...

File	Time
✓ 137Cs_2011-03.asc	2011/03
✓ 137Cs_2011-04.asc	2011/04
✓ 137Cs_2011-05.asc	2011/05
✓ 137Cs_2011-06.asc	2011/06
✓ 137Cs_2011-07.asc	2011/07
✓ 137Cs_2011-08.asc	2011/08
✓ 137Cs_2011-09.asc	2011/09

☐ Data is seasonal until 2100/01

Set file times

☐ Starting at 2011/03 spaced at 1 month(s)

☒ From file name (select which part): 137Cs_2011-03.asc year-month

Apply

Defaults OK Cancel

Figure 10: The interface to configure a specific external data connection (e.g, a time series of maps). Each time series has a name to uniquely define the series in the Ecopath with Ecosim user interface, an optional description, a variable that only serves to limit where the external data connection appears in the Ecopath with Ecosim user interface, and a series of time-tagged maps that, in this particular figure, reflects historical concentrations of ^{137}Cs in the water off of Fukushima, at monthly time steps, since March 2011.

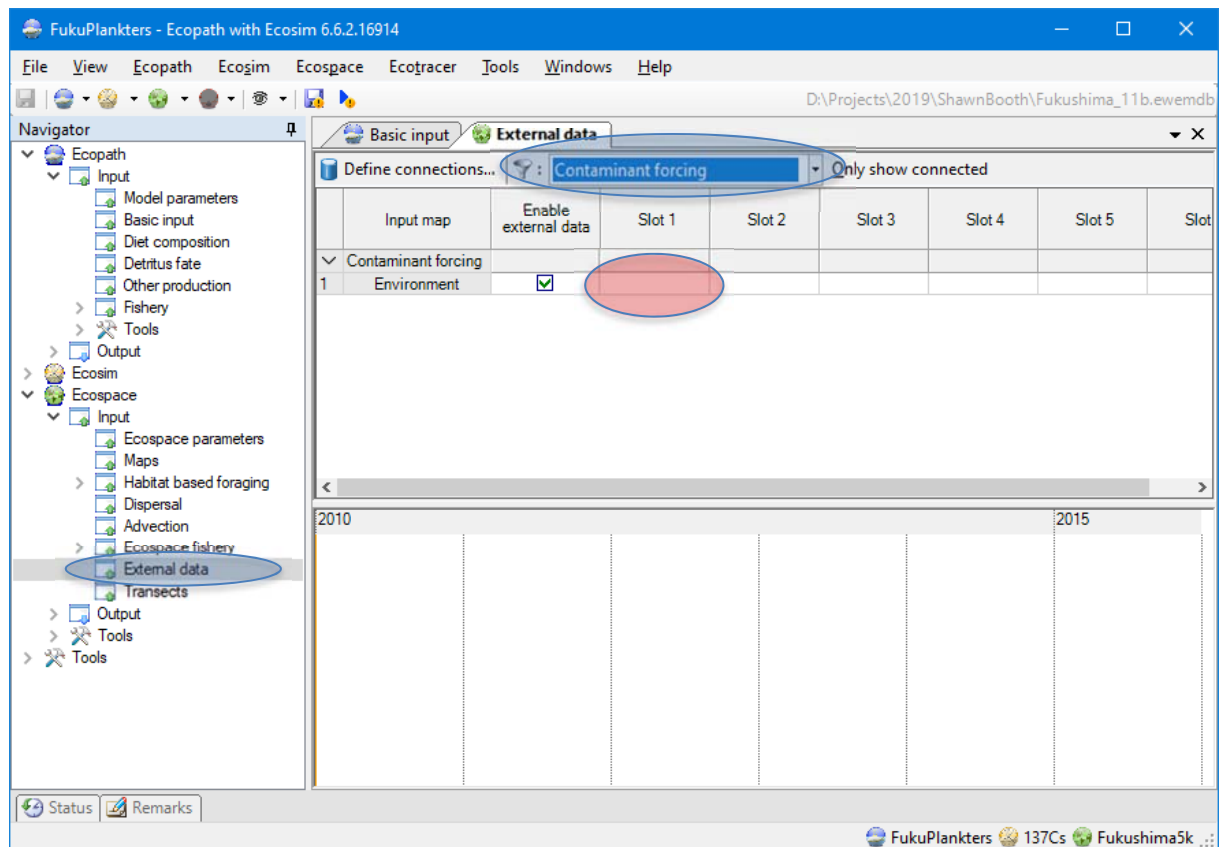


Figure 11: The External data interface where previously defined external data connections are applied to Ecospace layers.

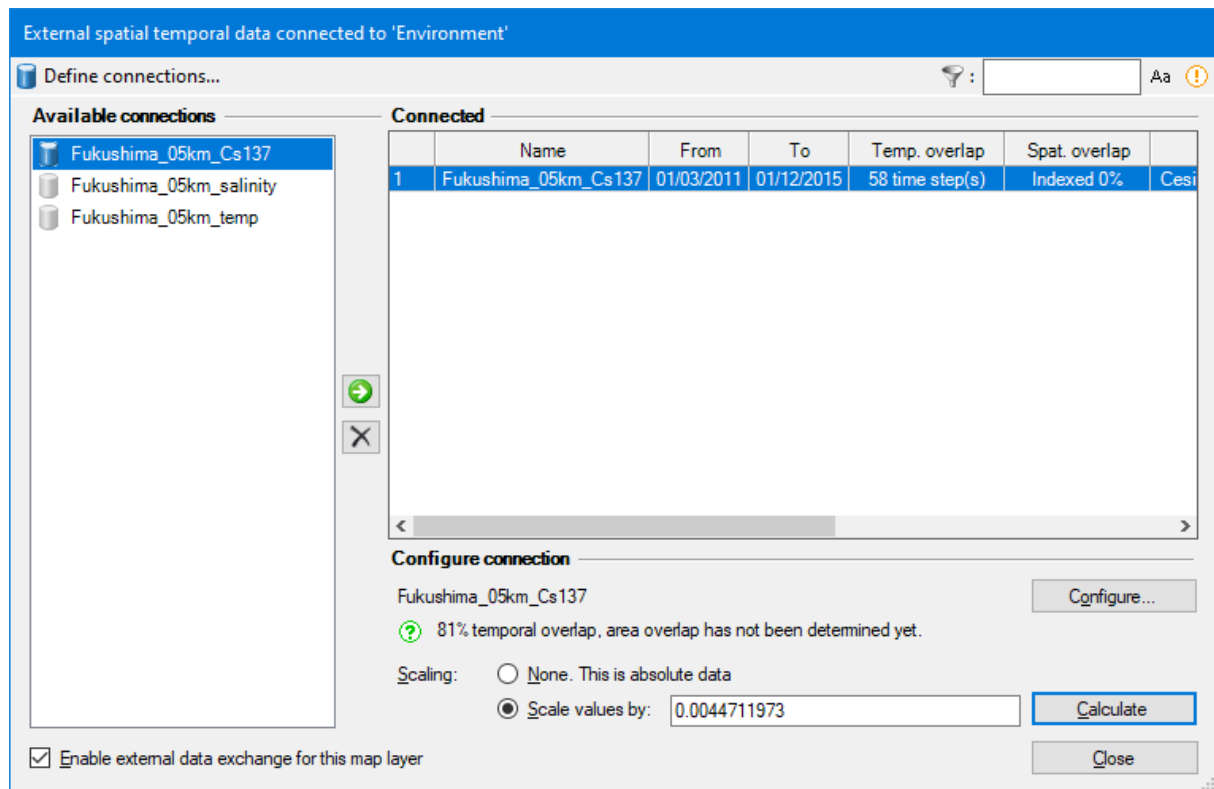


Figure 12: Configuring the integration of an external time series of maps into Ecospace, in this case to the environmental contaminants layer. Applied external time series are shown as "Connected". The selected connection can be configured, which includes entering a scaling factor to ensure that external data matches the magnitude of data currently in the Ecospace environmental contaminants layer. As described in Steenbeek et al. (2013), the Calculate option scales the average map mean for the first full year of external data to the map data loaded in Ecospace.